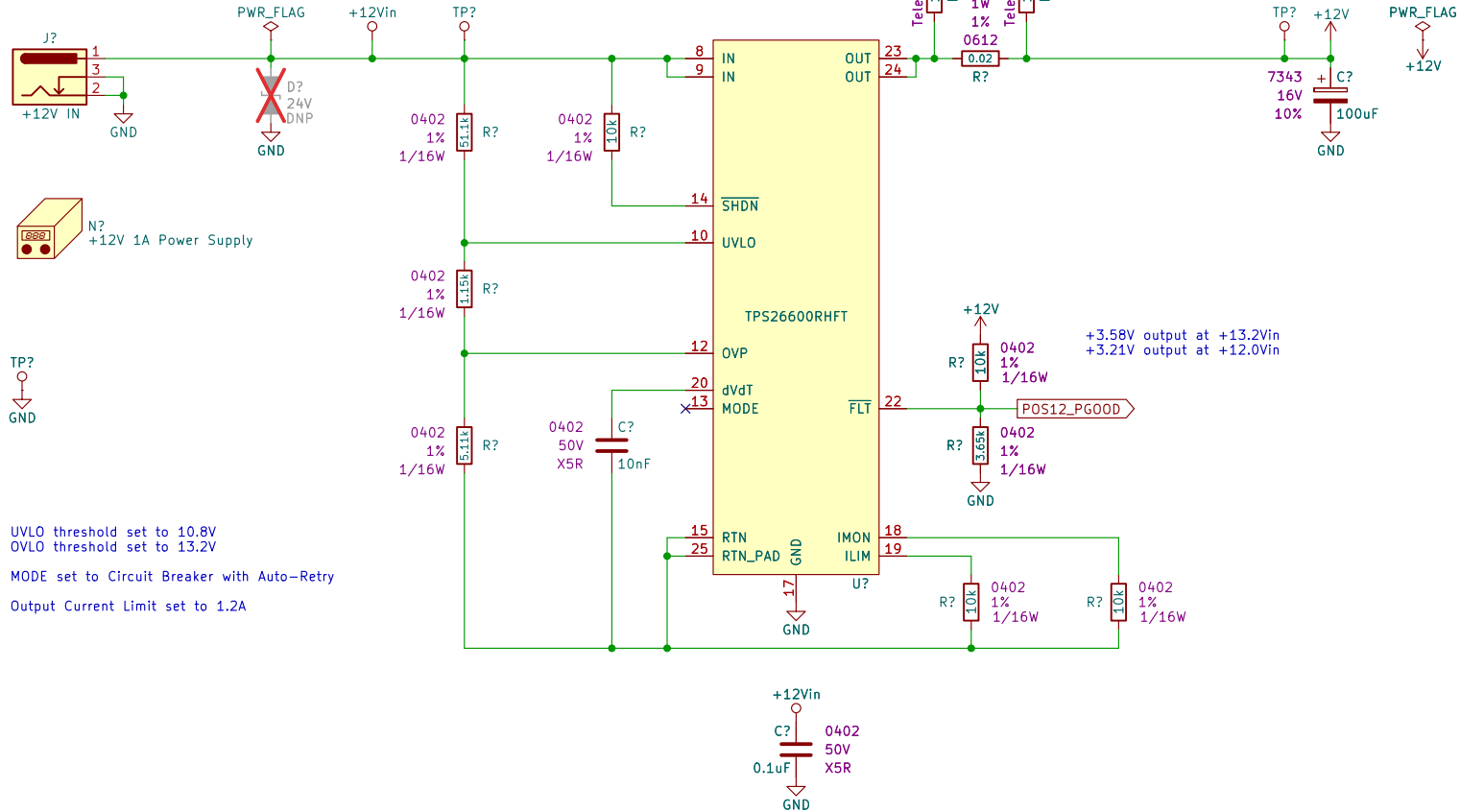


	1	2	3	4	5	
A	12V Power Input File: POS12_power_input.kicad_sch 12V Telemetry File: POS12_telemetry.kicad_sch 3.3V Power Supply File: POS3P3_power_supply.kicad_sch 3.3V Telemetry File: POS3P3_telemetry.kicad_sch dsPIC33 Programming File: dspic33_programming.kicad_sch dsPIC33 Power File: dspic33_power.kicad_sch dsPIC33 GPIO File: dspic33_gpio.kicad_sch USB UART Bridge File: usb_uart_bridge.kicad_sch	Platform ETC File: platform_etc.kicad_sch PGOOD LEDs File: pgood_leds.kicad_sch Status LEDs File: status_leds.kicad_sch RCA Signal Conditioning File: rca_signal_conditioning.kicad_sch Speaker Signal Conditioning File: speaker_signal_conditoning.kicad_sch Source Selection, PGAs File: source_selection_pgas.kicad_sch Anti-alias Filtering File: anti_alias_filtering.kicad_sch Audio Voltage References File: audio_voltage_references.kicad_sch				A
B						B
C						C
D						D
	1	2	3	4	5	

Sheet: / File: Audio_Visualiser_DSP.kicad_sch		
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Size: A	Date: 2023-12-25	Rev:
KiCad E.D.A. 10.0.3		Id: 1/17

TRY TO TALLY LOAD CURRENTS

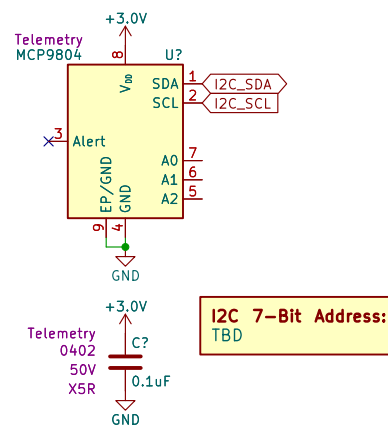
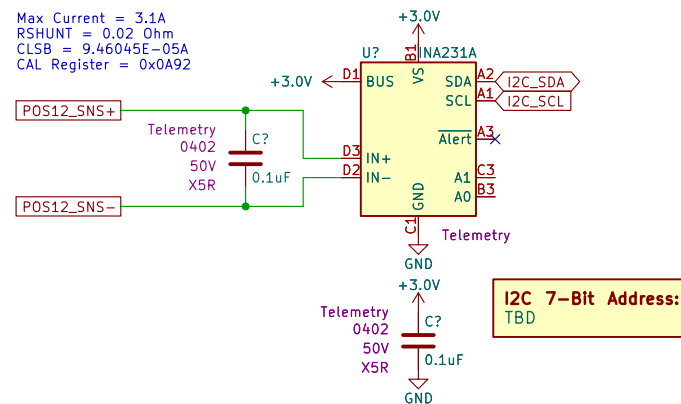


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Title:

Size: A Date: 2023-12-25
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Rev:
 Id: 2/17



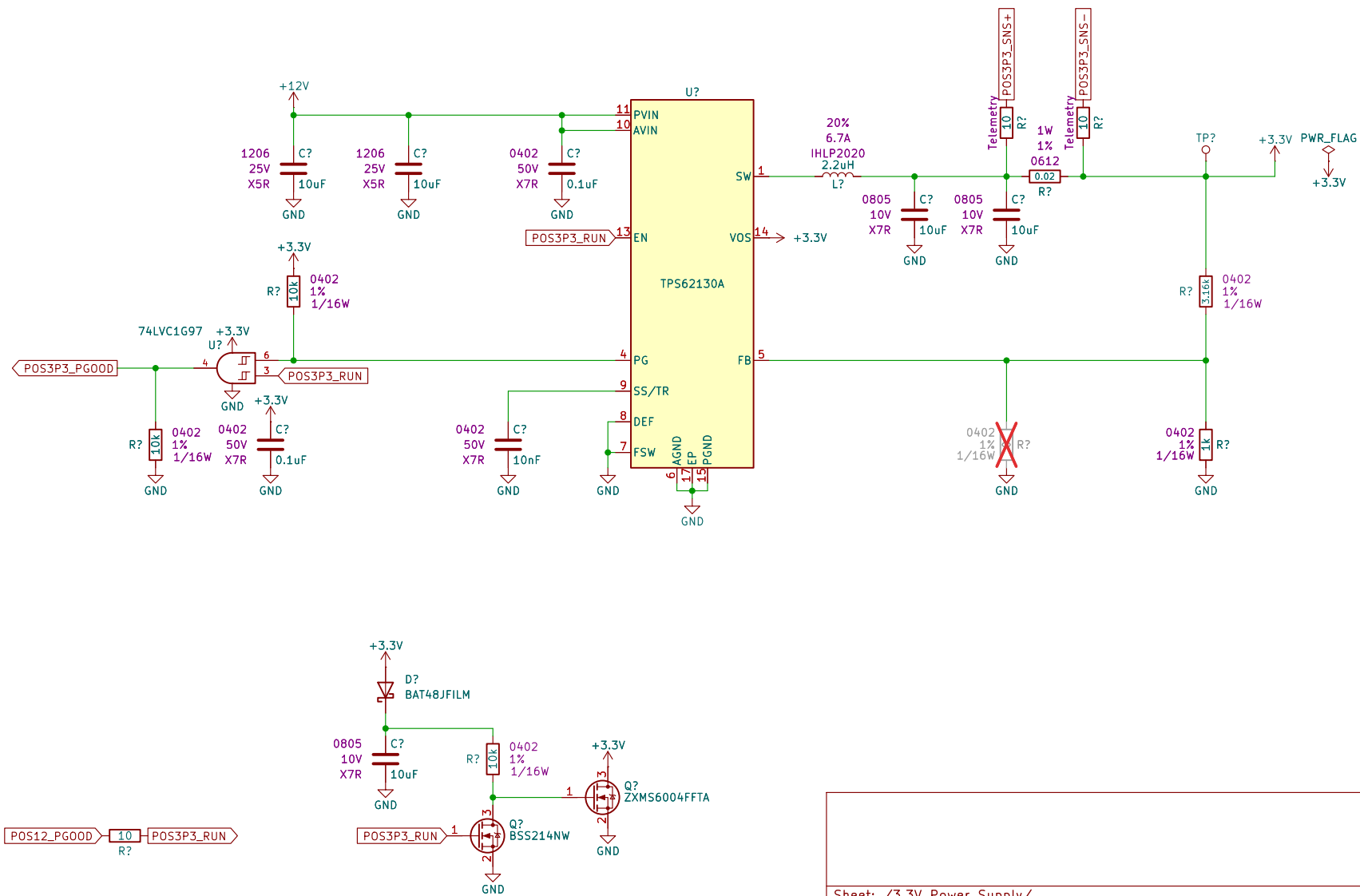
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Rev:
 Id: 3/17

TRY TO TALLY LOAD CURRENTS

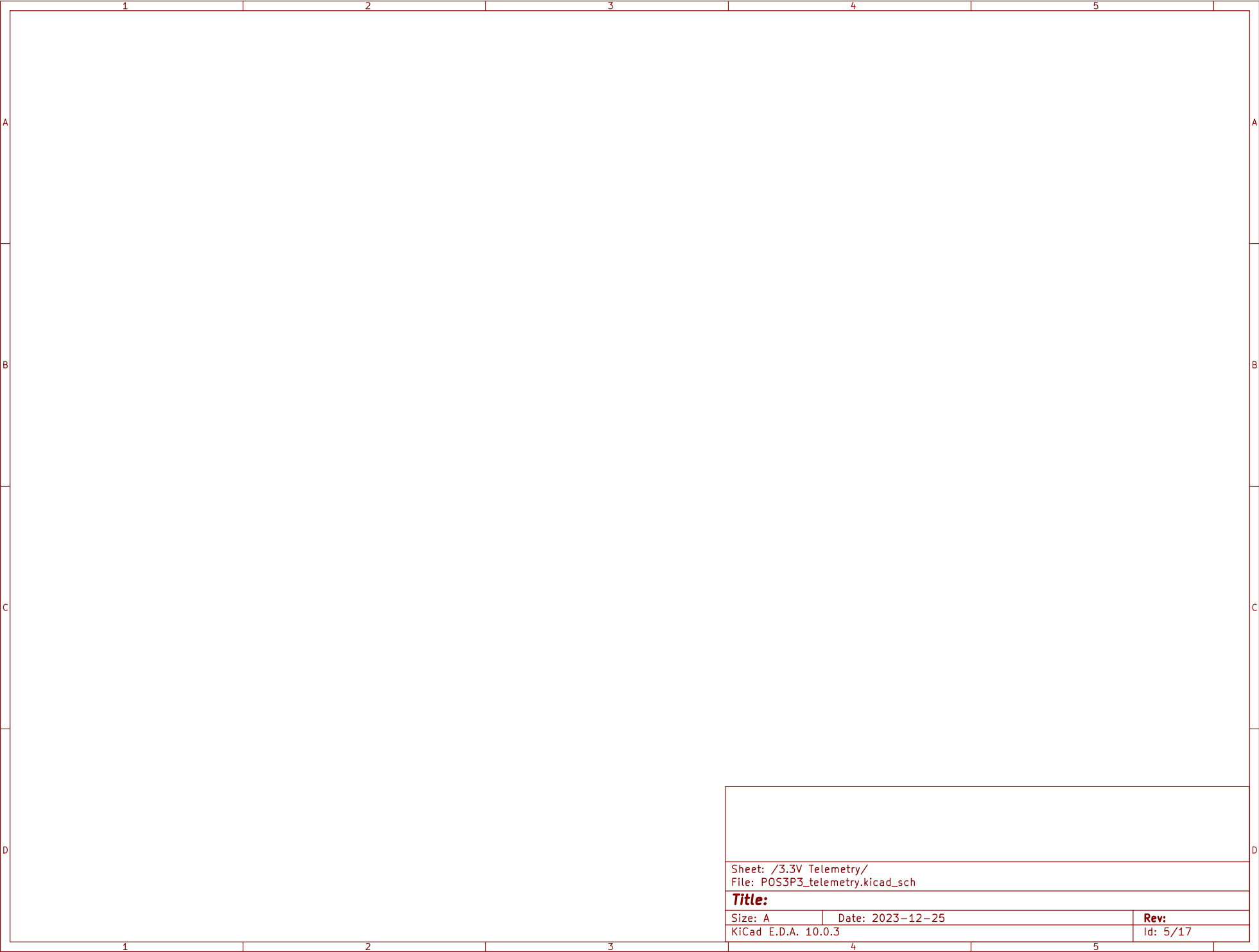


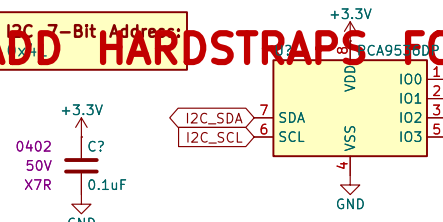
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File: POS3P3_power_supply.kicad_sch

Title:

Size: A Date: 2023-12-25
KiCad E.D.A. 10.0.3

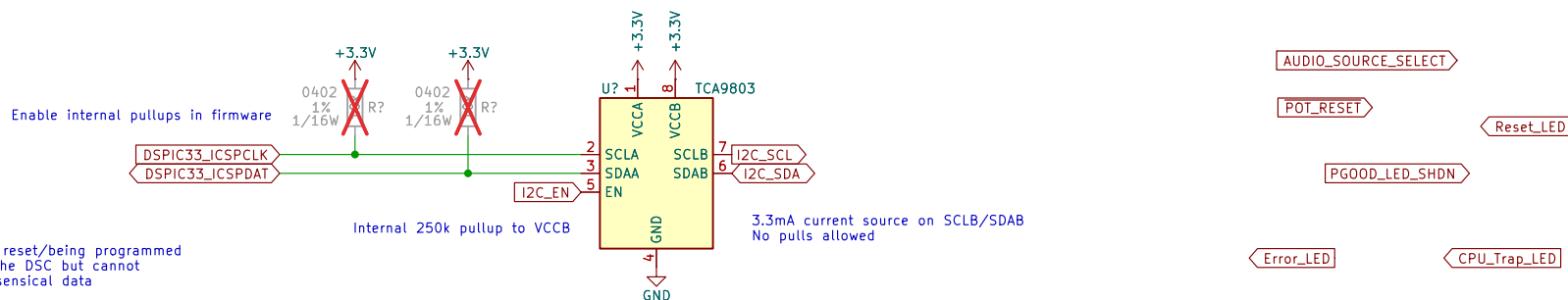
Rev:
Id: 4/17



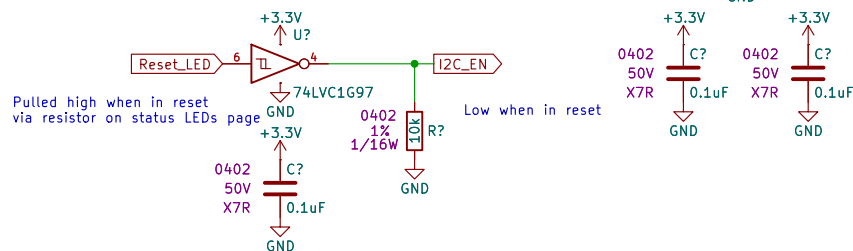


U?B

dsPIC33AK256MPS105



Isolate I2C signals when in reset/being programmed
so that ICSP can program the DSC but cannot
spook I2C devices with nonsensical data



Sheet: /dsPIC33 GPIO/
File: dspic33_gpio.kicad_sch

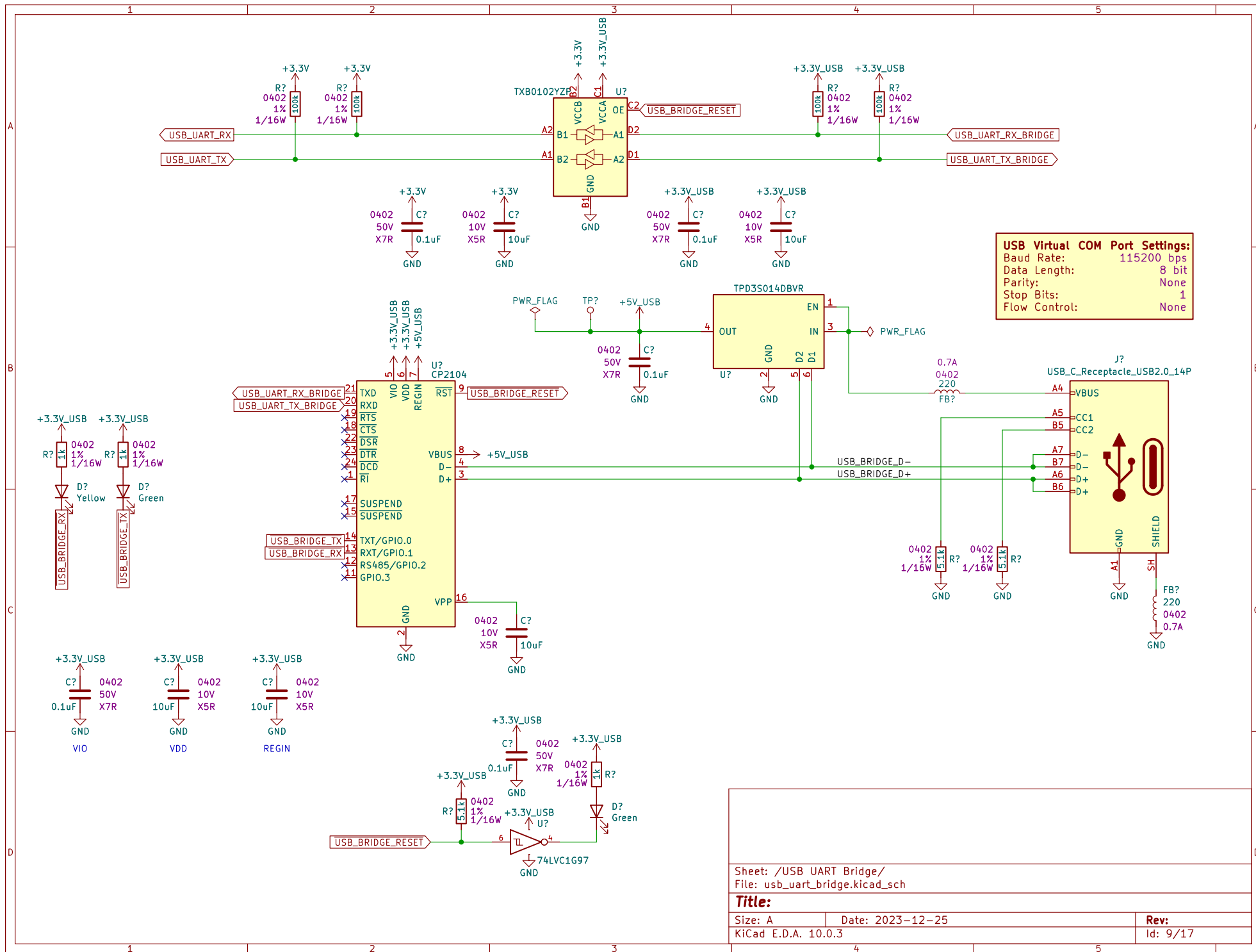
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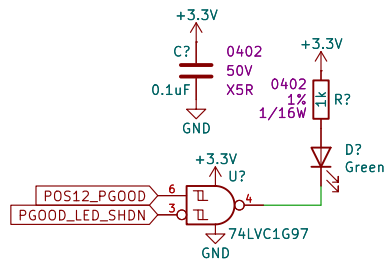
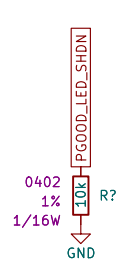
Rev:

Id: 8/17





Rev:
Id: 10/17



Sheet: /PGOOD LEDs/
File: pgood_leds.kicad_sch

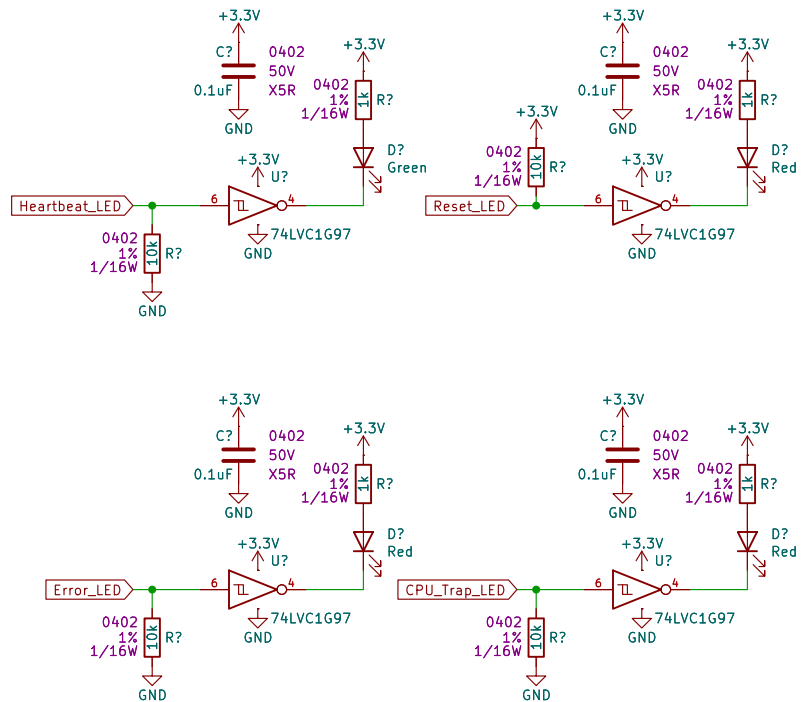
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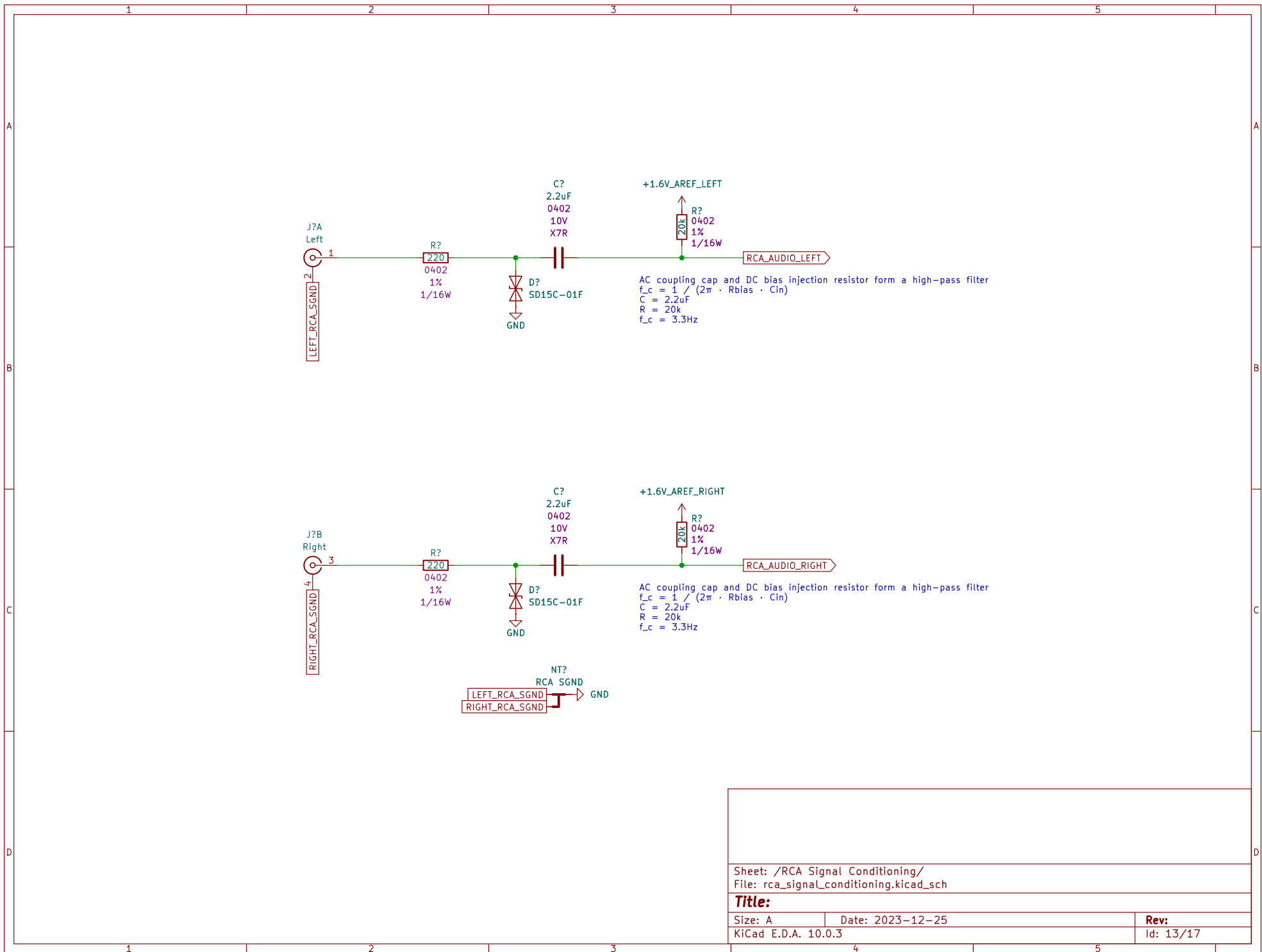


Sheet: /Status LEDs/
File: status_leds.kicad_sch

Title:

Size: A Date: 2023-12-25
KiCad E.D.A. 10.0.3

Rev:
Id: 12/17



Sheet: /RCA Signal Conditioning/
File: rca_signal_conditioning.kicad_sch

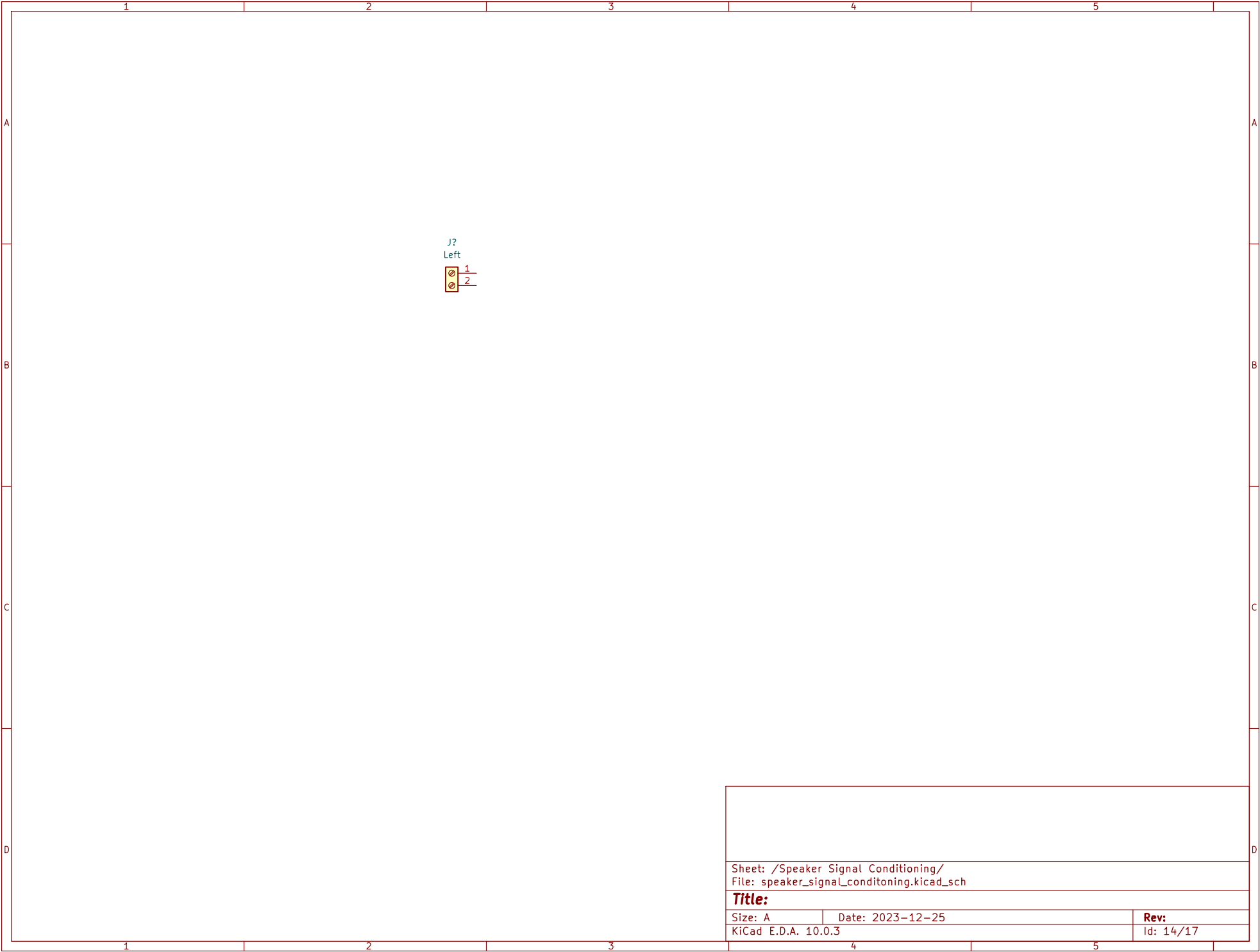
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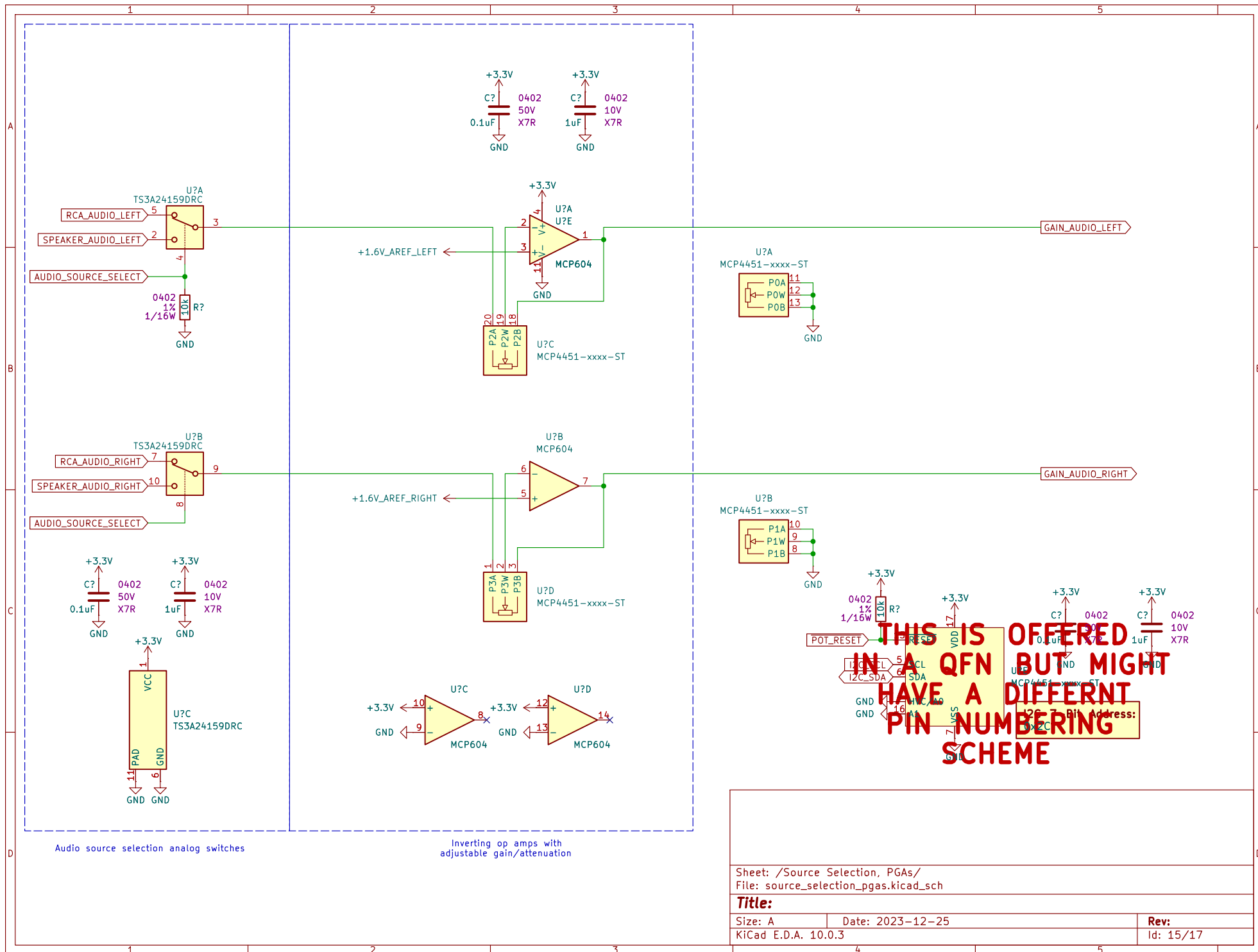
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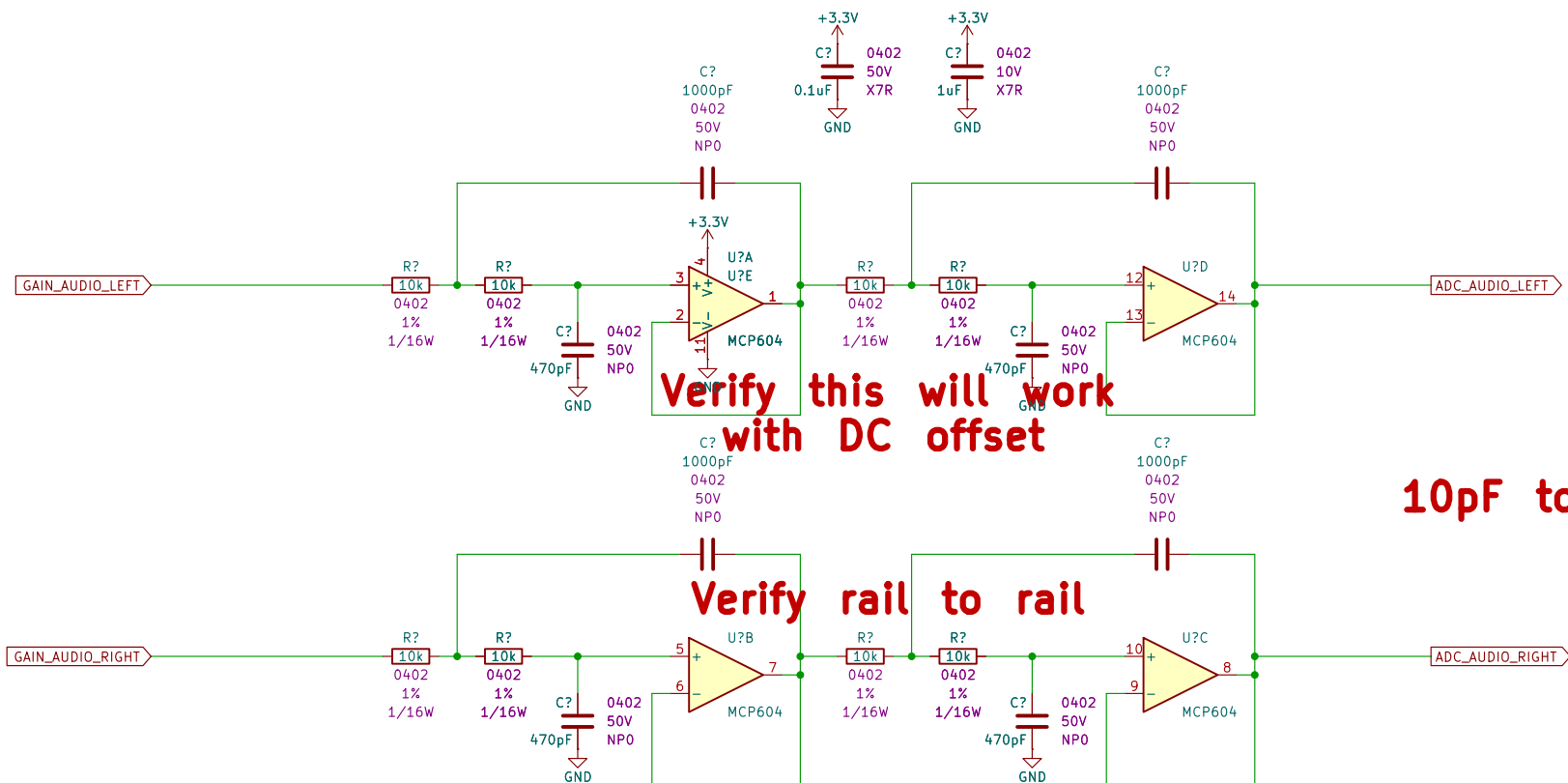
Rev:

Id: 13/17



Sheet: /Speaker Signal Conditioning/ File: speaker_signal_conditoning.kicad_sch		
Title:		
Size: A	Date: 2023-12-25	Rev:
KiCad E.D.A. 10.0.3		Id: 14/17





4-pole Sallen-Key active anti-aliasing lowpass filters
 Unity gain
 $f_c = 23.2\text{kHz}$, $Q = 0.707$ each
 If sampling at 44.1kHz , nyquist is 22.05kHz
 Must filter hard just below this

Sheet: /Anti-alias Filtering/
 File: anti_alias_filtering.kicad_sch

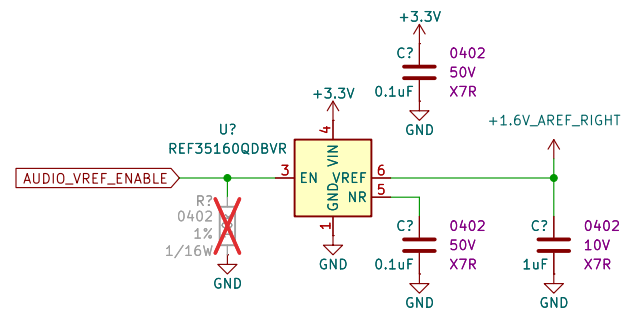
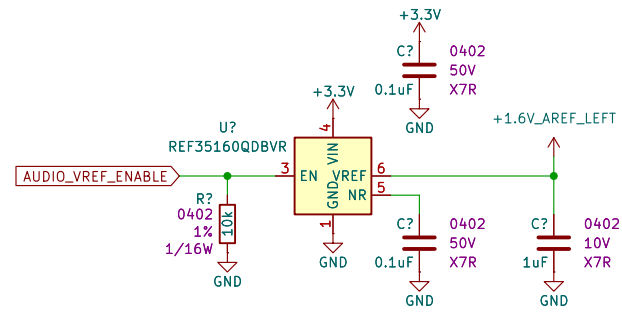
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Rev:

Id: 16/17



Sheet: /Audio Voltage References/
File: audio_voltage_references.kicad_sch

Title:

Size: A Date: 2023-12-25
KiCad E.D.A. 10.0.3

Rev:
Id: 17/17